1) a)

On line 2: He doesn't halve the -2 inside the bracket when completing the square.

On line 3: He subtracted 4 from the RHS rather than adding it and doesn't put a $\pm$ in front of the $\sqrt{-9}$

On line 4: He didn't square root the 9

1) b)

$z^2 - 2z + 5 = 0$

$(z-1)^2 - 1 = -5$

$z - 1 = \pm\sqrt{-4}$

$z = 1 \pm 2\mathrm{i}$

2) a)

$z^2 = -1$

$z = \pm\sqrt{-1}$

$z = \pm\mathrm{i}$

2) b)

$z^2 = -10$

$z = \pm\sqrt{-10}$

$z = \pm\sqrt{10}\mathrm{i}$

2) c)

$(z-1)^2 - 1 = -17$

$(z-1)^2 = -16$

$z - 1 = \pm\sqrt{-16}$

$z = 1 \pm 4\mathrm{i}$

2) d)

$4\left(z-\frac{1}{2}\right)^2-1=-17$

$4\left(z-\frac{1}{2}\right)^2=-16$

$\left(z-\frac{1}{2}\right)^2=-4$

$z-\frac{1}{2}=\pm\sqrt{-4}$

$z=\frac{1}{2}\pm 2\mathrm{i}$

2) e)

$2\left(z-\frac{1}{2}\right)^2-\frac{1}{2}=-1$

$2\left(z-\frac{1}{2}\right)^2=-\frac{1}{2}$

$\left(z-\frac{1}{2}\right)^2=-\frac{1}{4}$

$z-\frac{1}{2}=\pm\sqrt{-\frac{1}{4}}$

$z=\frac{1}{2}\pm\frac{1}{2}\mathrm{i}$

2) f)

$\left(z-\frac{1}{2}\right)^2-\frac{1}{4}=-3$

$\left(z-\frac{1}{2}\right)^2=-\frac{11}{4}$

$z-\frac{1}{2}=\pm\sqrt{-\frac{11}{4}}$

$z=\frac{1}{2}\pm\frac{1}{2}\sqrt{11}\mathrm{i}$

3) a)

$(z+5)^2=-7$

$z+5=\pm\sqrt{-7}$

$z=-5\pm\sqrt{7}\,\mathrm{i}$

3) b)

$(z-1)^2 = -12$

$z - 1 = \pm\sqrt{-12}$

$z = 1 \pm \sqrt{12}\,\mathrm{i}$

$z = 1 \pm 2\sqrt{3}\,\mathrm{i}$

3) c)

$2(z+4)^2 + 8 = 6$

$2(z+4)^2 = -2$

$(z+4)^2 = -1$

$z + 4 = \pm\sqrt{-1}$

$z = -4 \pm \mathrm{i}$

3) d)

$8(z+5)^2 - 3 = -9$

$8(z+5)^2 = -6$

$(z+5)^2 = -\frac{3}{4}$

$z + 5 = \pm\sqrt{-\frac{3}{4}}$

$z = -5 \pm \frac{\sqrt{3}}{2}\,\mathrm{i}$

4)

$z^4 + 2z^2 - 3 = 0$

$(z^2+1)^2 - 1 = 3$

$(z^2+1)^2 = 4$

$z^2 + 1 = \pm 2$

$z^2 = 1$ or $z^2 = -3$

$z = \pm 1$ or $z = \sqrt{-3} = \pm\sqrt{3}\mathrm{i}$

5) a)

$z^2 - (1 + \mathrm{i} + 1 - \mathrm{i})z + (1+\mathrm{i})(1-\mathrm{i})$

$= z^2 - z + \mathrm{i}z - z - \mathrm{i}z + 1 + \mathrm{i} - \mathrm{i} + 1$

$= z^2 - 2z$

5) b)

$z^2 - (3 - \mathrm{i} + 3 + \mathrm{i})z + (3 + \mathrm{i})(3 - \mathrm{i})$

$= z^2 - 3z + 3\mathrm{i}z - 3z - 3\mathrm{i}z + 9 + 3\mathrm{i} - 3\mathrm{i} + 1 = z^2 - 6z + 10$

5) c)

$z^2 - (1 + 2\mathrm{i} + 1 - 2\mathrm{i})z + (1 + 2\mathrm{i})(1 - 2\mathrm{i})$

$= z^2 - z + 2\mathrm{i}z - z - 2\mathrm{i}z + 1 + 2\mathrm{i} - 2\mathrm{i} + 4 = z^2 - 2z + 5$

5) d)

$z^2 - (-4 + 3\mathrm{i} - 4 - 3\mathrm{i})z + (-4 + 3\mathrm{i})(-4 - 3\mathrm{i})$

$= z^2 + 4z - 3\mathrm{i}z + 4z + 3\mathrm{i}z + 16 + 12\mathrm{i} - 12\mathrm{i} + 9 = z^2 + 8z + 25$

5) e)

$z^2 - (\mathrm{i} - \mathrm{i})z + (\mathrm{i})(-\mathrm{i})$

$= z^2 - \mathrm{i}z + \mathrm{i}z + 1 = z^2 + 1$

5) f)

$z^2 - (5\mathrm{i} - 5\mathrm{i})z + (5\mathrm{i})(-5\mathrm{i})$

$= z^2 - 5\mathrm{i}z + 5\mathrm{i}z + 25 = z^2 + 25$

5) g)

$z^2 - \left(-1 + \sqrt{2}\mathrm{i} - 1 - \sqrt{2}\mathrm{i}\right)z + \left(-1 + \sqrt{2}\mathrm{i}\right)\left(-1 - \sqrt{2}\mathrm{i}\right)$

$= z^2 + z - \sqrt{2}\mathrm{i}z + z + \sqrt{2}\mathrm{i}z + 1 - \sqrt{2}\mathrm{i} + \sqrt{2}\mathrm{i} + 2 = z^2 + 2z + 3$

6) a)

$3 + 8\mathrm{i}$

6) b)

$\left(z - (3 + 8\mathrm{i})\right)\left(z - (3 - 8\mathrm{i})\right) = z^2 - (3 + 8\mathrm{i})z - (3 - 8\mathrm{i})z + (3 + 8\mathrm{i})(3 - 8\mathrm{i})$

$= z^2 - 3z - 8\mathrm{i}z - 3z + 8\mathrm{i}z + 9 + 24\mathrm{i} - 24\mathrm{i} + 64 = z^2 - 6z + 73$

So $p = -6$ and $q = 73$

7) a)

$-\frac{1}{2} + 4\mathrm{i}$

7) b)

$\left(z-\left(-\frac{1}{2}-4\mathrm{i}\right)\right)\left(z-\left(-\frac{1}{2}+4\mathrm{i}\right)\right)=z^2-\left(-\frac{1}{2}-4\mathrm{i}\right)z-\left(-\frac{1}{2}+4\mathrm{i}\right)z+\left(-\frac{1}{2}-4\mathrm{i}\right)\left(-\frac{1}{2}+4\mathrm{i}\right)$

$=z^2+\frac{1}{2}z+4\mathrm{i}z+\frac{1}{2}z-4\mathrm{i}z+\frac{1}{4}-2\mathrm{i}+2\mathrm{i}+16=z^2+z+\frac{65}{4}$

So $p=1$ and $q=\frac{65}{4}$

8) a)

$a-b\mathrm{i}$

8) b)

$\big(z-(a+b\mathrm{i})\big)\big(z-(a-b\mathrm{i})\big)=z^2-(a+b\mathrm{i})z-(a-b\mathrm{i})z+(a+b\mathrm{i})(a-b\mathrm{i})$

$=z^2-az-b\mathrm{i}z-az+b\mathrm{i}z+a^2+ab\mathrm{i}-ab\mathrm{i}+b^2=z^2-2az+a^2+b^2$

So $p=-2a$ and $q=a^2+b^2$

9) a)

$z^2-4z+7=0$

$(z-2)^2-4=-7$

$z-2=\pm\sqrt{-3}$

$z=2\pm\sqrt{3}\mathrm{i}$

9) b)

$\left|2+\sqrt{3}\mathrm{i}\right|=\sqrt{4+3}=\sqrt{7}$

So $p=\sqrt{7}$

9) c)

Area is 7π units^2

9) d)

$w=-2+\sqrt{3}\mathrm{i}$ or $w=-2-\sqrt{3}\mathrm{i}$

9) e)

$\left(w-\left(-2+\sqrt{3}\mathrm{i}\right)\right)\left(w-\left(-2-\sqrt{3}\mathrm{i}\right)\right)=w^2+4w+\left(-2+\sqrt{3}\mathrm{i}\right)\left(-2-\sqrt{3}\mathrm{i}\right)$

$=w^2+4w+7$

$a=4,\ b=7$

10)

Distinct complex roots: $b^2-4ac<0$

$k^2-4\times4<0$

$(k-4)(k+4)<0$

$\therefore -4<k<4$

11)

$az^2+c=0$

$z^2=-\frac{c}{a}$

$z=\pm\sqrt{-\frac{c}{a}}$

$\therefore$ it only has purely imaginary roots

12)

$az^2+bz=0$

$z(az+b)=0$

The roots are $z=0$ and $z=-\frac{b}{a}$ which are real